STACKS, SHEAVES ON ALGEBRAIC STACKS: LEMMA-MAP-FROM-QUASI-COHERENT

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2463-2504 of sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex; proof span is lines 2506-2542, the complete author proof (35 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 17 of queue_algebraic-stacks-sheaves.json (pb=1545, stmt 2463-2504, proof 2506-2542).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma equation-comparison-algebraic-spaces-modules: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 1721–1725. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma equation-comparison-modules: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 1382–1386. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-stackification-quasi-coherent: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 2197–2207. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma proposition-quasi-coherent: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 2351–2357. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 044Q: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 2542–2542. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:stacks-sheaves.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let (U, R, s, t, c) be a groupoid in algebraic spaces over S. Assume s, t are flat and locally of finite presentation. Let $\mathcal{X} = [U/R]$ be the quotient stack. Denote x the object of $\mathcal{X}$ over U. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_\mathcal{X}$-module, and let $\mathcal{H}$ be any object of $Mod(\mathcal{O}_\mathcal{X})$. The map*

$$\mathrm{Hom}_{\mathcal{O}_\mathcal{X}}(\mathcal{F}, \mathcal{H}) \longrightarrow \mathrm{Hom}_{\mathcal{O}_U}(x^*\mathcal{F}|_{U_{étale}}, x^*\mathcal{H}|_{U_{étale}}), \quad \phi \longmapsto x^*\phi|_{U_{étale}}$$

is injective and its image consists of exactly those $\varphi : x^*\mathcal{F}|_{U_{étale}} \to x^*\mathcal{H}|_{U_{étale}}$ which give rise to a commutative diagram

$$\begin{array}{ccccc} s^*_{small}(x^*\mathcal{F}|_{U_{étale}}) & \longrightarrow & (x \circ s)^*\mathcal{F}|_{R_{étale}} = (x \circ t)^*\mathcal{F}|_{R_{étale}} & \longleftarrow & t^*_{small}(x^*\mathcal{F}|_{U_{étale}}) \\ \downarrow{\scriptstyle s^*_{small}\varphi} & & & & \downarrow{\scriptstyle t^*_{small}\varphi} \\ s^*_{small}(x^*\mathcal{H}|_{U_{étale}}) & \longrightarrow & (x \circ s)^*\mathcal{H}|_{R_{étale}} = (x \circ t)^*\mathcal{H}|_{R_{étale}} & \longleftarrow & t^*_{small}(x^*\mathcal{H}|_{U_{étale}}) \end{array}$$

of modules on $R_{étale}$ where the horizontal arrows are the comparison maps (equation-comparison-algebraic-spaces-modules).

Proof. According to Lemma lemma-stackification-quasi-coherent the stackification map $[U/_pR] \to [U/R]$ (see Groupoids in Spaces, Definition 044Q) induces an equivalence of categories of quasi-coherent sheaves and of fppf $\mathcal{O}$-modules. Thus it suffices to prove the lemma with $\mathcal{X} = [U/_pR]$. By Proposition proposition-quasi-coherent and its proof there exists a quasi-coherent module $(\mathcal{G}, \alpha)$ on (U, R, s, t, c) such that $\mathcal{F}$ is given by the rule $\mathcal{F}(T, u) = \Gamma(T, u^*\mathcal{G})$. In particular $x^*\mathcal{F}|_{U_{étale}} = \mathcal{G}$ and it is clear that the map of the statement of the lemma is injective. Moreover, given a map $\varphi : \mathcal{G} \to x^*\mathcal{H}|_{U_{étale}}$ and given any object $y = (T, u)$ of $[U/_pR]$ we can consider the map

$$\mathcal{F}(y) = \Gamma(T, u^*\mathcal{G}) \xrightarrow{u^*_{small}\varphi} \Gamma(T, u^*_{small}x^*\mathcal{H}|_{U_{étale}}) \to \Gamma(T, y^*\mathcal{H}|_{T_{étale}}) = \mathcal{H}(y)$$

where the second arrow is the comparison map (equation-comparison-modules) for the sheaf $\mathcal{H}$. This assignment is compatible with the restriction mappings of the sheaves $\mathcal{F}$ and $\mathcal{G}$ for morphisms of $[U/_pR]$ if the cocycle condition of the lemma is satisfied. Proof omitted. Hint: the restriction maps of $\mathcal{F}$ are made explicit in terms of $(\mathcal{G}, \alpha)$ in the proof of Proposition proposition-quasi-coherent. □